Key Constraints in SQL SERVER

We will discuss:

1. What is a Constraint
2. All Key Constraints
3. Primary Key
4. Unique Key
5. Composite Key
6. Candidate Key
7. Alternate Key
8. Foreign Key
9. Referential Integrity (very important)
10. Practical Examples
11. Interview Points

1️⃣ What is a Constraint?

A **constraint** is a rule enforced at the database level to ensure **data accuracy, consistency, and integrity**.

Without constraints:

- Duplicate data occurs
- Invalid data enters
- Relationships break

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2 Types of Key Constraints

Constraint	Purpose
PRIMARY KEY	Uniquely identifies record
UNIQUE KEY	Ensures uniqueness
FOREIGN KEY	Maintains relationship
COMPOSITE KEY	Multiple columns as key
CANDIDATE KEY	Possible primary key
ALTERNATE KEY	Candidate key not chosen as PK

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3 PRIMARY KEY (PK)

Definition

A Primary Key:

- Uniquely identifies each row
- Cannot be NULL
- Only one per table
- Automatically creates clustered index (by default in SQL Server)

Example

```
CREATE TABLE Students (
    StudentId INT PRIMARY KEY,
    Name VARCHAR(100),
    Email VARCHAR(100)
);
```

Rules:

- No duplicate StudentId
- No NULL allowed

Why Primary Key is Important?

- Identity of record
- Used for relationships
- Ensures entity integrity

4️⃣ UNIQUE KEY

Definition

Ensures column values are unique but:

- Can allow one NULL (SQL Server behavior)
- A table can have multiple UNIQUE constraints

Example

```
CREATE TABLE Employees (
    EmployeeId INT PRIMARY KEY,
    Email VARCHAR(100) UNIQUE
);
```

Here:

- EmployeeId = PK
- Email = UNIQUE KEY

Difference Between PK and UNIQUE

Primary Key	Unique Key
Only one per table	Multiple allowed
Cannot be NULL	Can allow one NULL
Main identifier	Secondary uniqueness

5️⃣ COMPOSITE KEY

A key made of multiple columns.

Example

```
CREATE TABLE StudentCourses (
    StudentId INT,
    CourseId INT,
    PRIMARY KEY (StudentId, CourseId)
);
```

Here:

- Combination must be unique
- Individually they can repeat
- Together they cannot repeat

Used in:

- Many-to-many relationships

6️⃣ CANDIDATE KEY

A column (or set of columns) that can uniquely identify a row.

Example:

Students table:

| StudentId | Email | Aadhaar |

All 3 are unique.

Possible Candidate Keys:

- StudentId
- Email
- Aadhaar

One is chosen as Primary Key.
Others become Alternate Keys.

7️⃣ ALTERNATE KEY

Candidate key not selected as Primary Key.

Example:

Email UNIQUE

Email becomes Alternate Key.

8️⃣ FOREIGN KEY (FK)

Definition

A Foreign Key:

- Creates relationship between two tables
- Enforces referential integrity
- Must match Primary Key or Unique Key of parent table

Parent Table

```
CREATE TABLE Customers (
   CustomerId INT PRIMARY KEY,
   Name VARCHAR(100)
);
```

Child Table

```
CREATE TABLE Orders (
   OrderId INT PRIMARY KEY,
   CustomerId INT,
   FOREIGN KEY (CustomerId)
      REFERENCES Customers(CustomerId)
);
```

9️⃣ Referential Integrity (Very Important)

What is Referential Integrity?

It ensures that:

1. You cannot insert child record without parent
2. You cannot delete parent if child exists (unless rule defined)
3. You cannot update parent key if child exists (unless cascade defined)

Referential Integrity Rules

Rule 1: Insert Rule

❌ Not Allowed:

```
INSERT INTO Orders VALUES (1, 100);
```

If CustomerId 100 does not exist.

Rule 2: Delete Rule

If customer has orders:

DELETE FROM Customers WHERE CustomerId = 1;

❌ Error (default behavior)

Rule 3: Update Rule

UPDATE Customers SET CustomerId = 10 WHERE CustomerId = 1;

❌ Error if child exists.

🔟 Foreign Key Options (Very Important)

When defining FK, we can control behavior:

ON DELETE Options

Option	Behavior
NO ACTION (default)	Prevent delete
CASCADE	Delete child rows
SET NULL	Set FK to NULL
SET DEFAULT	Set default value

Example with CASCADE

FOREIGN KEY (CustomerId)

REFERENCES Customers(CustomerId)

ON DELETE CASCADE

ON UPDATE CASCADE

Now:

- Delete parent → child deleted
- Update parent → child updated

1️⃣1️⃣ Real-Life Example for Students

Think of:

- Department → Employees
- Customer → Orders
- Order → OrderItems

If:

- Parent deleted
- What should happen to child?

This is business logic decision.

1️⃣2️⃣ Entity Integrity vs Referential Integrity

Integrity Type	Enforced By	Meaning
Entity Integrity	Primary Key	No duplicate identity
Referential Integrity	Foreign Key	Relationship consistency

1️⃣3️⃣ Interview-Focused Summary

Be able to answer:

✓ Difference between Primary Key and Unique
✓ Can Foreign Key reference Unique key? (Yes)
✓ What is Composite Key?
✓ What is Candidate Key?

✓ What happens if parent is deleted?
✓ Difference between CASCADE and SET NULL

1️⃣ 4️⃣ Best Practices (Industry Level)

- Primary key should be stable (do not update)
- Avoid unnecessary cascade deletes in financial systems
- Always define FK explicitly (do not rely only on app layer)

Final Concept Map

Primary Key → Identity of table
Unique Key → Additional uniqueness
Composite Key → Multi-column identity
Foreign Key → Relationship
Referential Integrity → Data consistency